

Impact of Prompt Engineering Training on Learning Outcomes in an Engineering Course

Section A

No Training

retained n = 13

Section B

Moderate

retained n = 13

Section C

Extended

retained n = 19

Same instructor, lectures, and four assignments

Each assignment required ChatGPT use and submitted logs; training intensity varied by section.

Pre & Post Survey

55 pre and 45 post responses
9 dimensions
pre/post composites

Grades

Midterm & final grades
45 retained participants

Prompt Quality

Instructor-rated 1–5
144 scored prompt observations

Analysis: Participant-level contrasts and correlations; prompt trajectory model; adjusted HC3 models; FDR correction

Key Finding 1

Extended training showed higher prompt quality
C vs pooled A+B:
 $\Delta = 0.64$,
 $g = 0.77$, $p = 0.014$

Key Finding 2

Better prompts correlated with higher final grades
Pearson $r = 0.451$
95% CI [0.181, 0.657], $n = 45$

Key Finding 3

Higher pre-test locus of control predicted lower prompt quality
std. $\beta = -0.558$
FDR $p = 0.0015$